

13/8/26

CO3 - Tree Traversal Challenges

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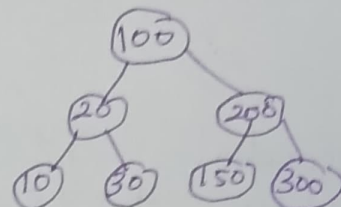
Reg no:- 192524372

Course:- CSA0811

Course ~~code~~: Data Structure

Write Traversal Sequence for following trees:-

Q-1:-



Preorder traversal (NLR)

100, 20, 10, 30, 200, 150, 300

Inorder traversal (LNR)

10, 20, 30, 100, 150, 200, 300

Postorder traversal (LRN)

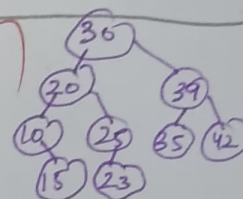
10, 30, 20, 150, 300, 200, 100

Q-2

Preorder:- 30, 20, 10, 15, 25, 23, 39, 35, 42

Inorder:- 10, 15, 20, 23, 25, 30, 35, 39, 42

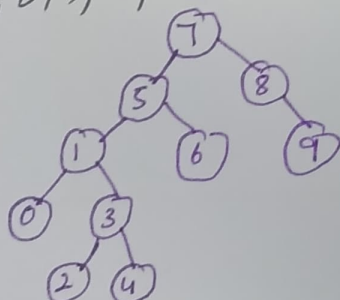
Postorder:- 15, 10, 23, 25, 20, 35, 42, 39, 30



Q-3:- Binary Search Tree.

7, 5, 1, 8, 3, 6, 0, 9, 4, 2

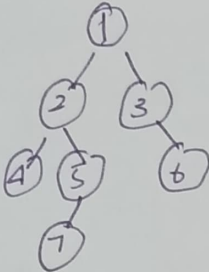
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Inorden: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Preorden: 7, 5, 1, 0, 3, 2, 4, 6, 8, 9

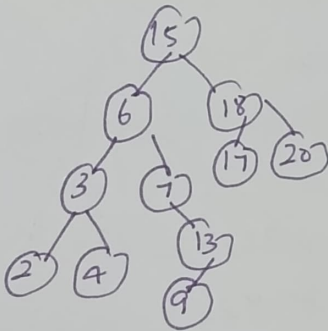
Postorden: 0, 2, 4, 3, 1, 6, 5, 9, 8, 7



Preorden: 1, 2, 4, 5, 7, 3, 6

Inorden: 4, 2, 7, 5, 1, 3, 6

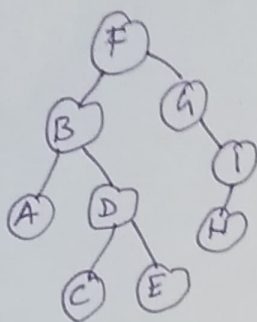
Postorden: 4, 7, 5, 2, 6, 3, 1



Preorden: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Postorden: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

Inorden: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20



Traversal :

A, B, C, D, E, F, G, H, I

Option - 3

Inorder: D, B, E, F, A, G, H, I

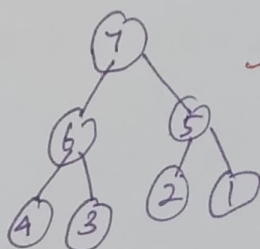
Preorder: A, B, D, E, F, C, G, H

What is the Postorder:

The answer is:

D, F, E, B, H, G, C, A

Option - 3

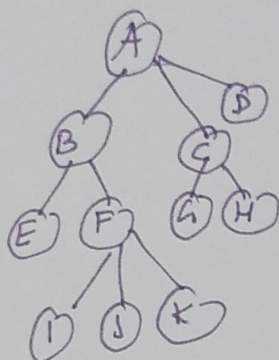


Inorden: - 4, 6, 3, 7, 2, 5, 1

Preorden: - 7, 6, 4, 3, 5, 2, 3

Postorden: - 4, 3, 6, 2, 1, 5, 7

10)



Preorden: visit root first

A, B, E, F, I, J, K, G, H, D

Postorden: visit children first

E, I, J, K, F, B, G, H, C, D, A

Level orden:

A, B, C, D, E, F, G, H, I, J, K.